Module 6: Test and Commission of a Solar PV System

Module Outline



Unit 6.1
Tools and Accessories for Testing



Unit 6.2
Overall System Inspection



Unit 6.3
Testing of Solar Array



Unit 6.4
Wire and Earthing Continuity Tests



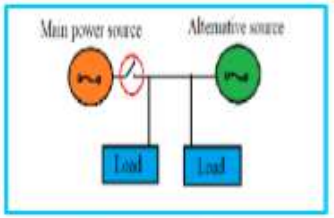
Unit 6.5
Testing of Charge Controller



Unit 6.6
Testing of Batteries



Unit 6.7
Start-up the System



Unit 6.8
Unintentional Islanding Functionality Tests



Unit 6.9
Inspection Record Sheet

Unit 6.1: Tools and accessories for testing

6.1.1. Tools to be Used

You need tools:

- To take measurements.
- To carry out electrical and mechanical parameters.
- To test the functionality of the Solar PV systems.

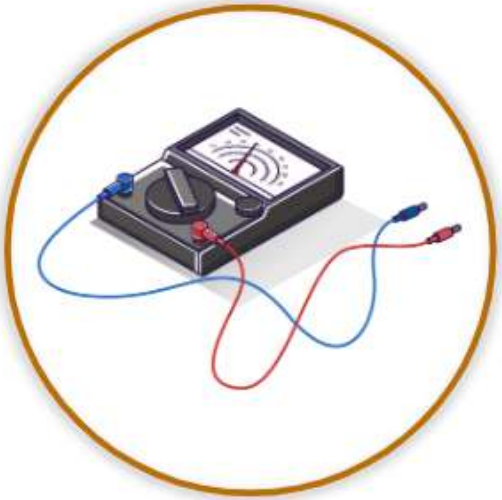


List of tools

1. Safety equipment
2. Multimeter
3. A clamp meter
4. A Hydrometer
5. A screwdriver set
6. Nut Driver
7. A Crimping Tool Set
8. A sun pathfinder or solar sitting device
9.Compass
10. Angle Gauge
11. Battery Safety Accessories
12. Battery Water Filler
13. Battery Maintenance Kit
14. A measuring tape/ digital distance meter
15. A camera
16. A notebook
17. Testing and commissioning worksheets
18. A flashlight

Safety Gear and First Aid Box





Multi-meter



Clamp Meter



Hydrometer



Screwdrivers



Nut Drivers



Crimping Tool Sets



Sun Pathfinder



Compass



Angle Gauge



Battery Safety Accessories



Battery Water Filler



Battery Maintenance Kit



Measuring Tape/ Digital Measuring Meter



Camera



Notebooks and Pencils



Test and Commission Reports

Points to Remember:

- Use relevant worksheets for inspection and testing.
- Conduct the tests on a clear sunny day.
- To rectify issues with the wiring before switching on the inverter and preventing damage, commissioning should begin at the PV array.



Before commissioning a system, the installer should ensure:

- All strings are in segments to prevent accidental arcs (i.e. leave one of the module interconnections open).
- All fuses should be removed.
- All circuit breakers and isolators are in the 'off' position (including the AC and DC isolators at the inverter).
- Inverter should be turned off.



Unit 6.2: Overall System Inspection

Location Assessment of Array, Equipment, and Batteries

Let's begin with:

- Assessment of Array Location
- Checking Equipment Location, and
- Checking Battery Location



Location Assessment of Array, Equipment, and Batteries



Assess Array Location

- PV array shall be free from shadow in all days of the year – use sun path finder.
- There should be safe and convenient access for array maintenance and cleaning.
- Array shall be protected from animals and vandalism.
- There should be ample space for airflow on the back side of the array.
- The aesthetic of the building or premises shall be taken into account.
- Array should not be located far from the charge controller/ inverter/ battery.



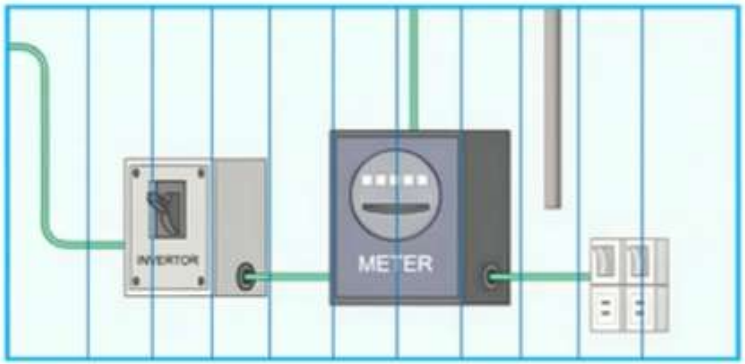
This is the correct site for installation.



Incorrect site selection for array installation

Check Equipment Location

Ensure all controls, power conditioning equipment, and instruments are installed in a controlled access area.



Check if electrical disconnect switches or circuit breakers are installed in such a way that access is controlled.



The access to the inverter is controlled



Inverter and control equipment installed in a control room

Check Battery Location

- Batteries should be in a dry location.
- Batteries should not be in contact with cold surfaces such as concrete.
- The battery should be enclosed in a container or box or room that allows for proper ventilation
- Batteries should have safe and easy access for maintenance and replacement.







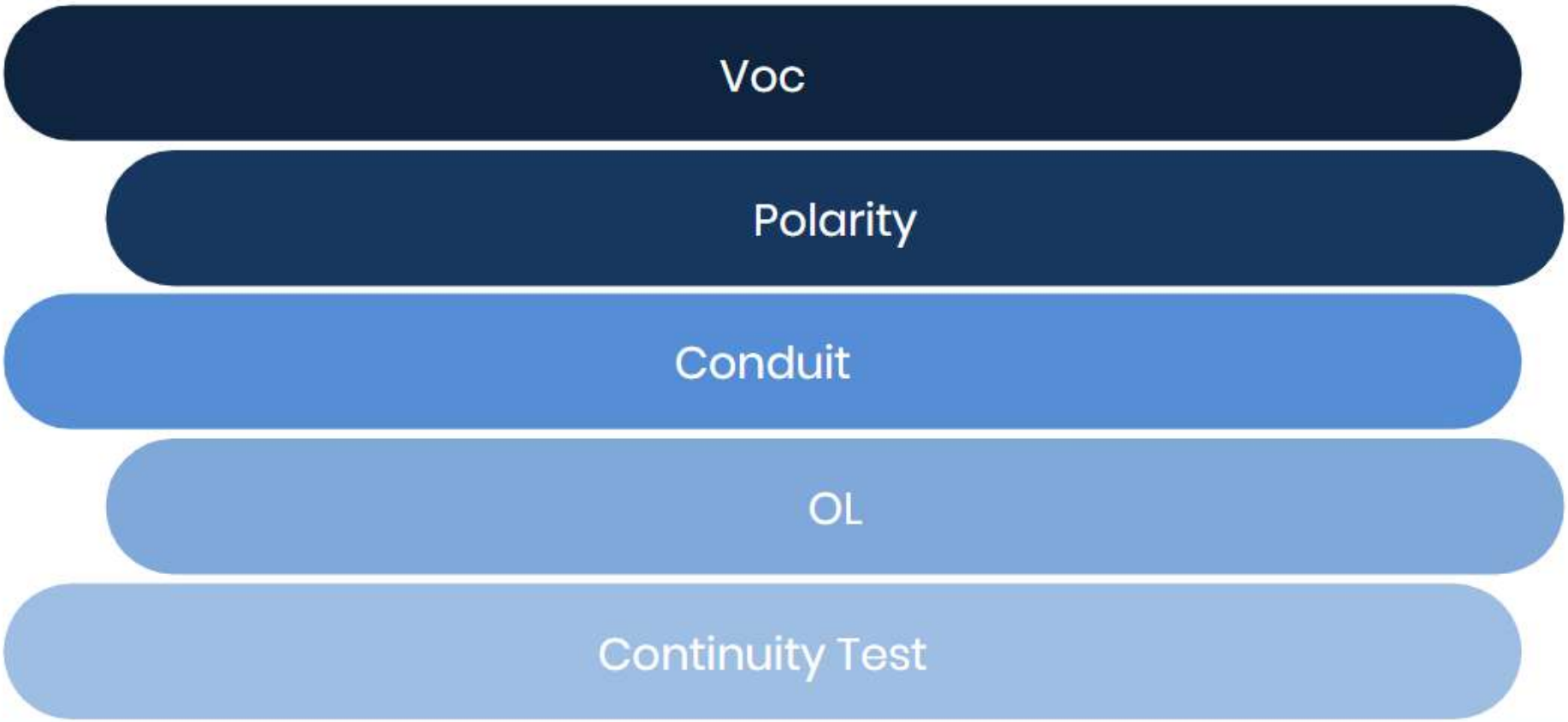
Batteries should always be kept in a box, container or rack



Location of batteries in a well-ventilated room

Unit 6.3: Testing of Solar Array

PV-pedia



PV-pedia

Voc

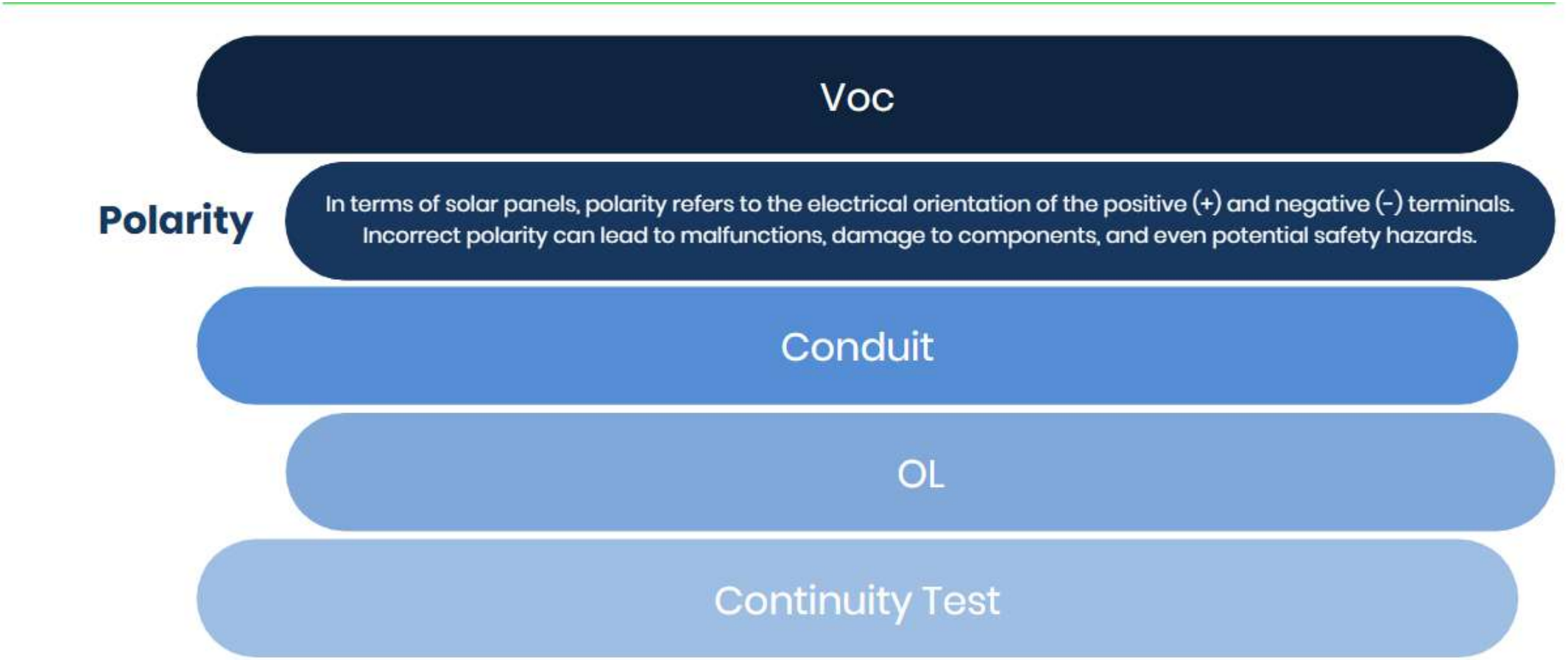
The open circuit voltage is the maximum voltage that the solar panel can produce with no load on it .
It is measured with a multimeter across the open ends of the wires attached to the panel).
If two or more panels are wired in series it will be Voc of panel 1 + Voc of panel 2, etc.

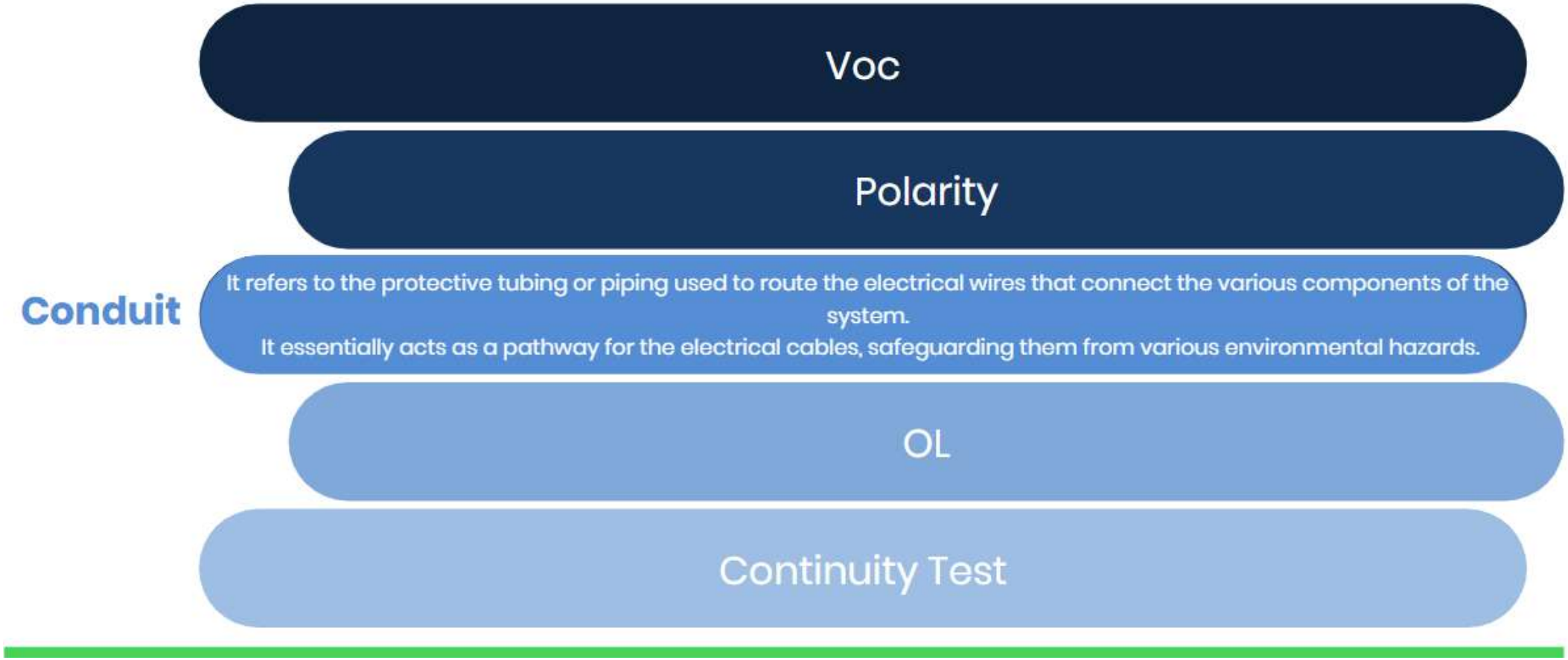
Polarity

Conduit

OL

Continuity Test


Voc
Polarity
In terms of solar panels, polarity refers to the electrical orientation of the positive (+) and negative (-) terminals. Incorrect polarity can lead to malfunctions, damage to components, and even potential safety hazards.
Conduit
OL
Continuity Test


Voc
Polarity
Conduit
It refers to the protective tubing or piping used to route the electrical wires that connect the various components of the system.
It essentially acts as a pathway for the electrical cables, safeguarding them from various environmental hazards.
OL
Continuity Test

Voc

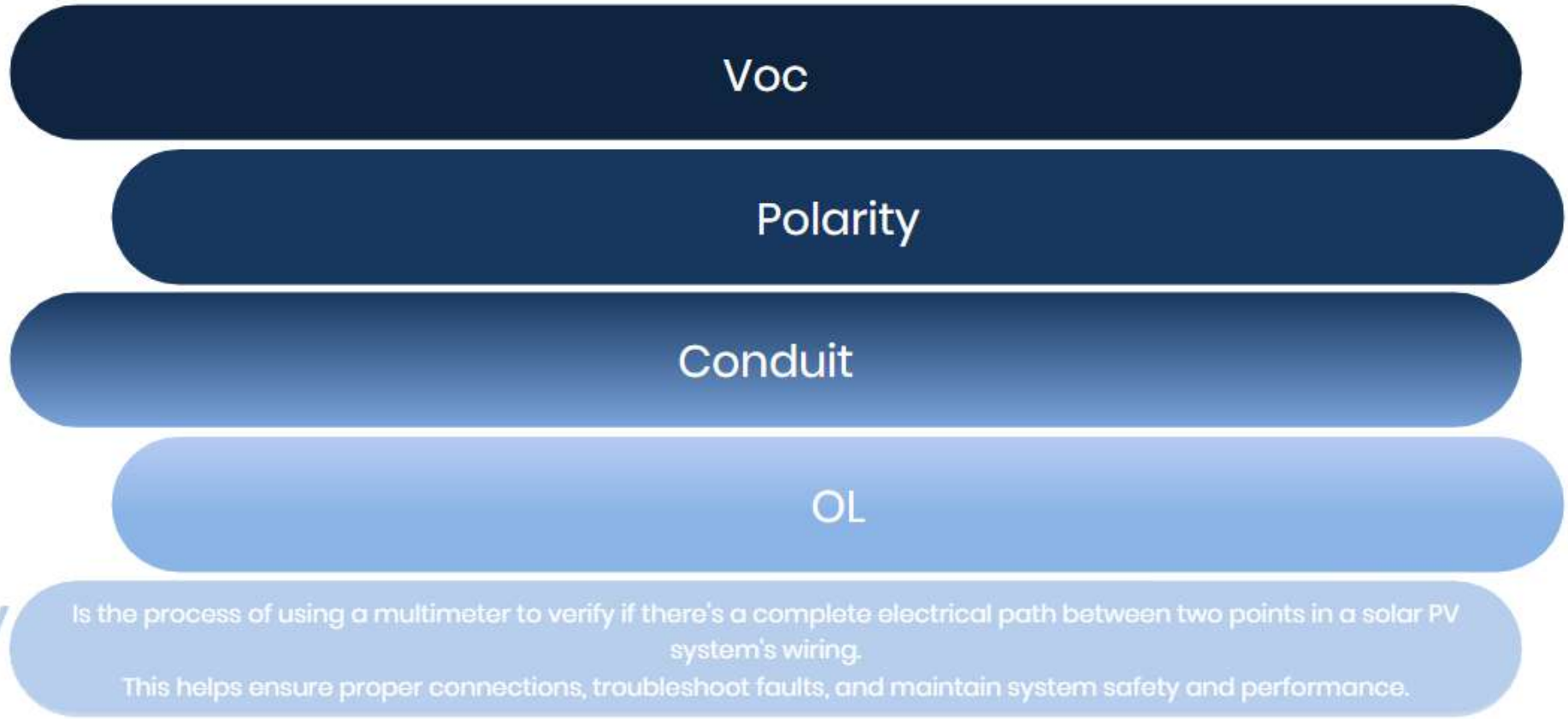
Polarity

Conduit

OL

This stands for "Open Loop" and signifies that the multimeter cannot detect any current flowing between the probes because the circuit is incomplete.
It essentially tells you there's an infinitely high resistance between the probes.

Continuity Test



Continuity Test

Is the process of using a multimeter to verify if there's a complete electrical path between two points in a solar PV system's wiring.
This helps ensure proper connections, troubleshoot faults, and maintain system safety and performance.

6.3.1 Overview of Testing Methods and Parameters

A well-functioning solar PV system requires verification at various stages to ensure optimal performance, safety, and longevity.



Steps for Conducting Tests and Evaluation



Types of Tests to Determine the Efficiency of the Solar PV System



Testing of photovoltaic array



Continuity Test



Testing Voltage and Polarity

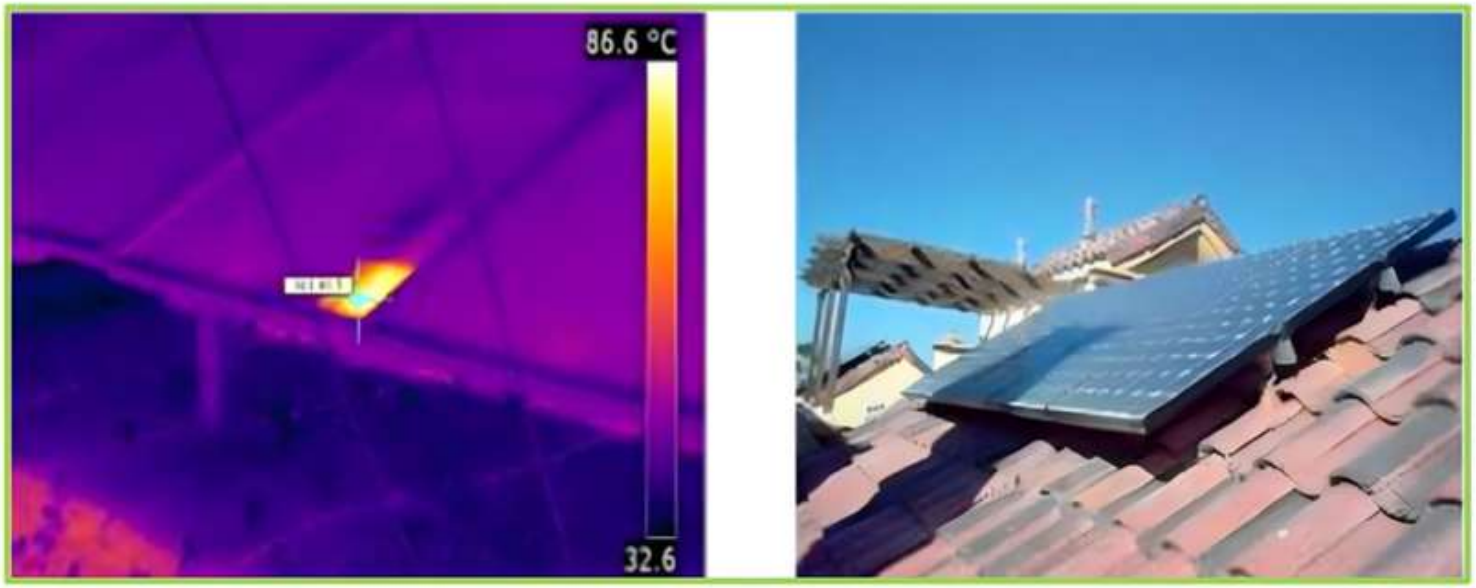
Steps for Conducting Tests of Photovoltaic Array

Step 1: Check the physical condition of the photovoltaic array for any physical damage. Module frames should be straight with no corrosion.

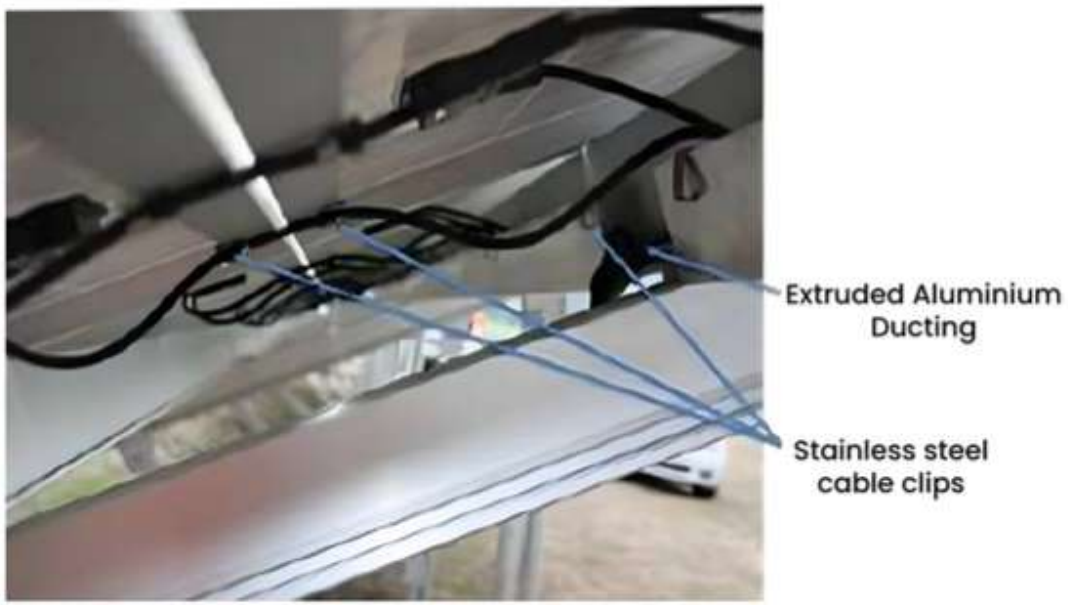


Steps for Conducting Tests of Photovoltaic Array

Step 2: All modules should be unshaded throughout the day. 'Spot' shading of a few cells or modules in the entire array must be eliminated.



Step 3: Check all the mounting hardware for loose fasteners or connections to the mounting surface.



Step 4: Conduit and connections must all be tight and undamaged.



Step 5: Visually check all conduit and wire insulation for damage.





Step 6: Check junction boxes if loose or broken.



Step 7: Make sure that wiring is secured, by gently but firmly pulling on all connections.



Step 8: Check if array wirings are NOT readily accessible to un-authorised personnel.

Step 9: Check all strings for continuity, appropriate voltage, and, lastly, polarity.





Step 10: If the voltages of the strings are not within 5% of Voc (Open Circuit Voltage) or are not what you would expect for the number of modules and weather conditions, you need to investigate this before proceeding further.

Continuity Test

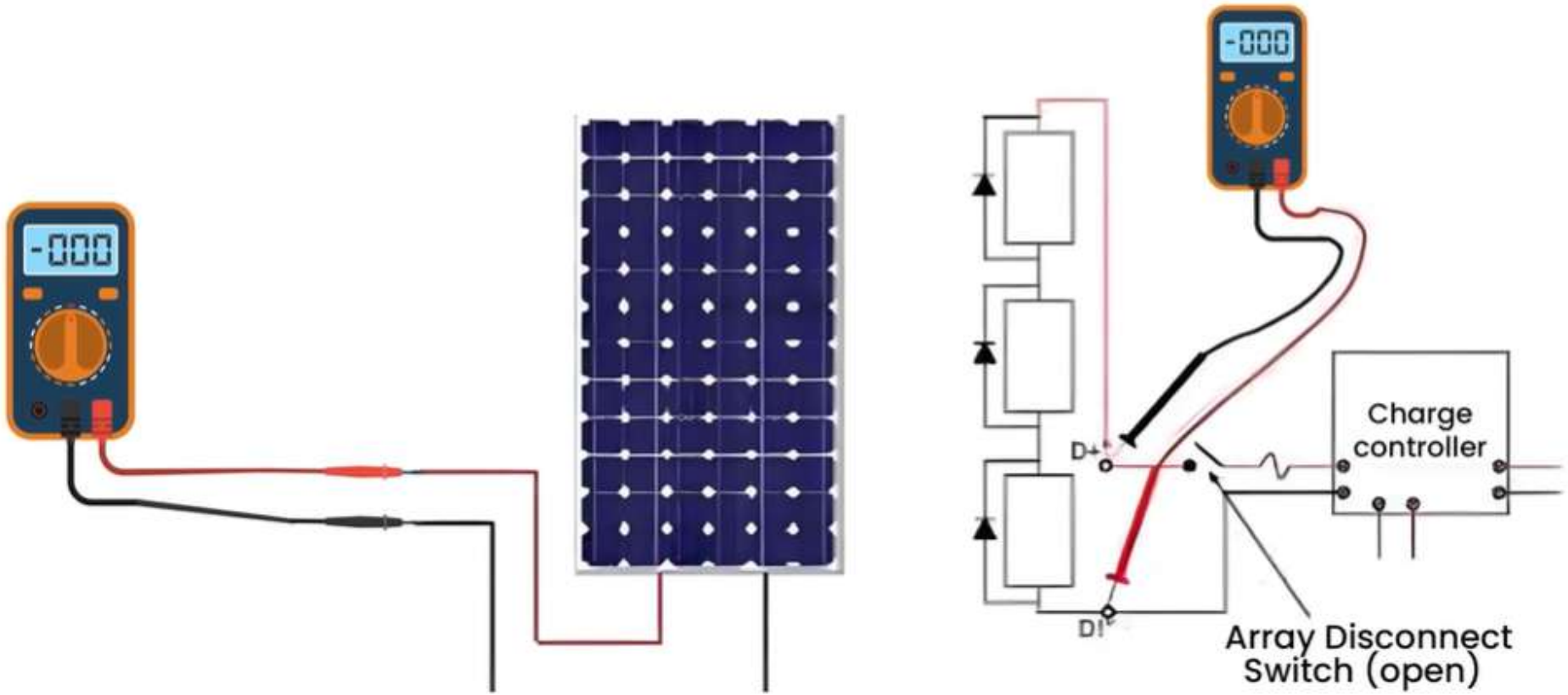


Continuity Test

1. **Step 1**: Turn the dial to Continuity Test mode With the test probes separated, the multimeter's display will show Open Loop (OL) and Ohms (Ω).
2. **Step 2**: First insert the black test lead into the COM jack. Then insert the red lead into the V Ω jack.
3. **Step 3**: Connect the test leads across the tested string in open circuit condition.
4. **Step 4**: The digital multimeter (DMM) beeps if a complete path (continuity) is detected. If the string is not connected the DMM will not beep.
5. **Step 5**: Switch OFF the multimeter.

Voltage and Polarity Test




-000
-000
Charge
controller
D+
D-
Array Disconnect
Switch (open)

Always remember, that loose connection and wrong polarity can cause fire in the PV system, completely damaging the system and property.



Testing Open Circuit Voltage and Polarity for Multiple Modules




1st and 2nd Modules Connection in Series


Ready Strings for 1 Kilowatt System


85.2 Volts


SCGJ
SKILL COUNCIL FOR GREEN JOBS
FLUKE
85.3

Unit 6.4: Wire and earthing continuity test

Conducting Wire and Earthing Tests

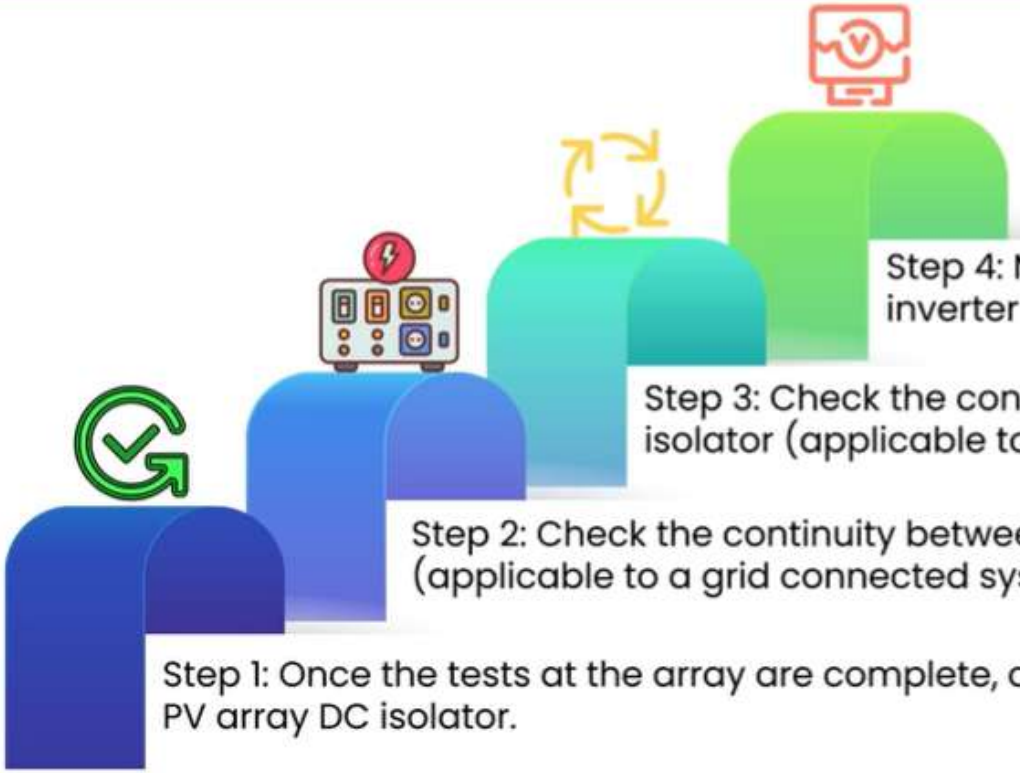
1. Continuity test confirms electrical path between the equipment, grounding rods, and other grounding points.
2. Continuity tests can identify loose or corroded connections within the earthing system.
3. The continuity test helps ensure that the resistance between the equipment and the earth meets the required safety standards.



Tests for Earthing and Lightning Protection



Continuity Tests



Step 4: Measure the voltage of the grid on the output of the inverter AC isolator(applicable to a grid connected system).

Step 3: Check the continuity between the kWh meter and the inverter AC isolator (applicable to a grid connected system).

Step 2: Check the continuity between the inverter and the inverter AC isolator. (applicable to a grid connected system).

Step 1: Once the tests at the array are complete, continuity should be confirmed from the array to the PV array DC isolator.

Steps Performed to Test Lightning and Earthing Protection

Step 1 — Turn off all the disconnect switches.

Step 2 — Use an ohmmeter to check the continuity of the entire grounding system.

Step 3 — Make sure that all module frames, metal conduit and connectors, junction boxes, and electrical components chassis are earth grounded.

Step 4



Using a DC voltmeter, check the polarity of all system components and wiring.

Step 5



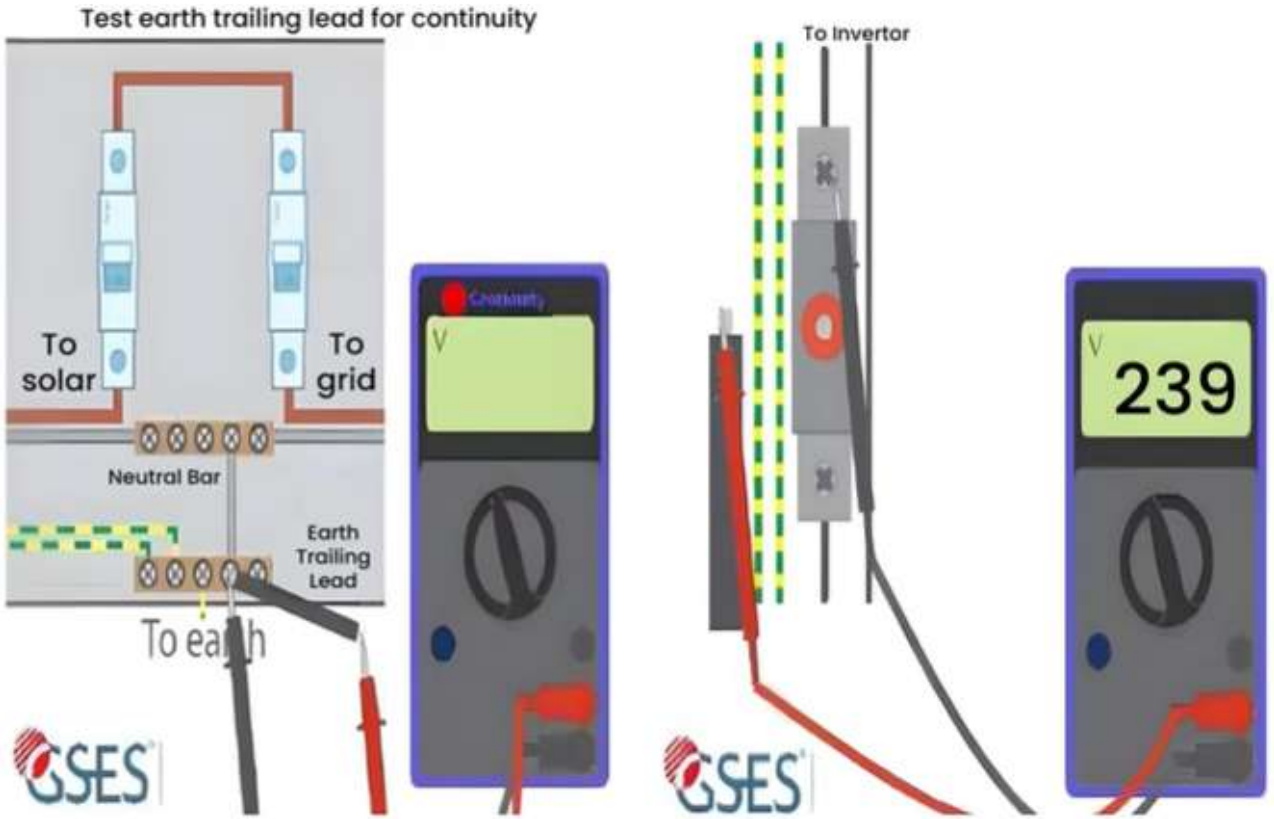
If plastic conduit is used, make sure a grounding wire has been run through it to provide continuous grounding.

Step 6



If a metal conduit is used, it functions as the ground conductor, where allowed by code. If not allowed by code, a grounding wire must be used.

Earth Continuity and Leakage Test



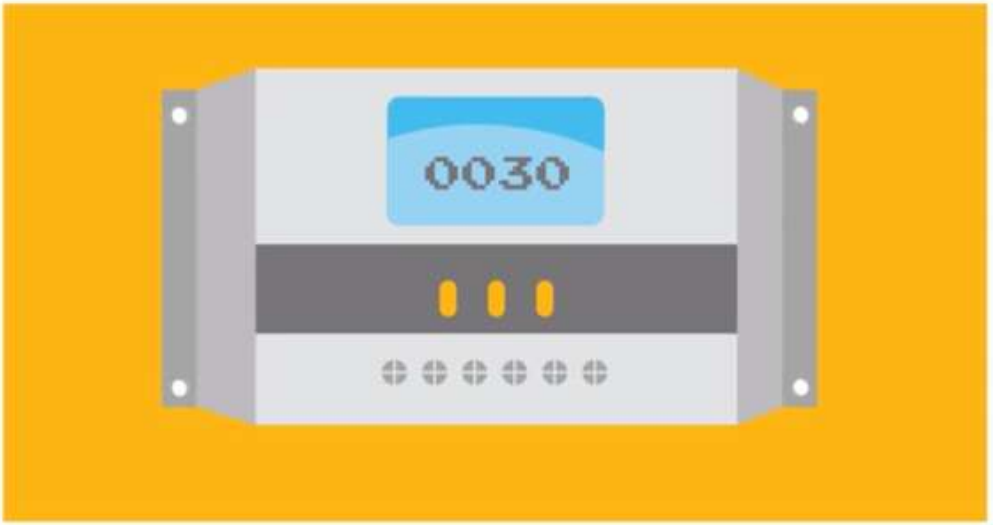
Checking Continuity of DCDB Wires



Cables are Continuous

Unit 6.5: Testing of Charge Controller

Steps to Test Charge Controller



Steps to Test Charge Controller



Step 1: Follow the manufacturer's instructions, if available, for the specific charge controller in the system



Step 2: Check all terminals and wires for loose, broken, corroded, or burnt connections or components



Step 3: Check all displays, LED indicators, and status monitoring systems are in operatio



Step 4: Check that overcharge protection and under charge protection of the charge controller are functioning correctly.



Step 5: If the charge controller is equipped with a temperature compensation device, check whether the temperature probe is functioning correctly.



Step 6: Make sure there are no loose strands of multi-strand wire, as they can short out on other terminals or wire

Testing of Charge Controller





Testing Procedure for Shunt Charge Controller

Step 1

Set the multimeter to the appropriate DC voltage range to measure the voltage between the array positive and array negative (terminals).





Measure the DC voltage between the battery positive and battery negative terminals on the controller. If the controller is operating properly, it should be between 13.5 and 14.5 volts per module in serie

Step 2

Testing Procedure for Shunt Charge Controller







Step 1: Disconnect all wiring from the controller, except the temperature compensation probe, if the controller has one. Set the power supply to zero volts

Step 2: Connect the power supply and DC voltmeter (Multimeter set at DC voltage) to the controller's + and - "array" input terminals.

Step 3: Watching the meter, slowly increase the power supply voltage until it is equal to the nominal voltage rating of the charge controller

Note:
You will need a DC portable adjustable power supply of a suitable voltage range to carry out this test



Step 4: Continue to increase the voltage until the meter reads one-half volt above the charge termination setting of the controller. The "charging" LED should go off. Record the charge termination voltage and compare it with the manufacturer's datasheet



Step 5: Turn the power supply voltage back to zero, then move the meter and power supply to the + and - "battery" terminals on the charge controller. .



Step 6: Slowly increase the Voltage. At first, the low voltage disconnect LED may be off

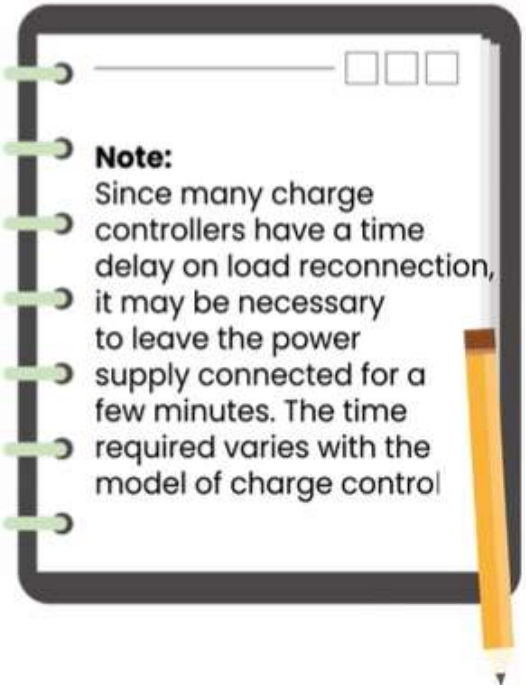
Testing Procedure for Series Charge Controllers

Step 7 : Once you supply enough voltage to operate the controller, but are still below the low voltage disconnect setting, the LED should be on.



Step 8: When the voltage is higher than the disconnect setting, the LED should go off. The voltage at which the LED comes on is the low battery reconnect voltage and should be recorded and compared with the manufacturer's datasheet.





Testing Procedure of Pulse Charger

Step 1 Turn the power supply voltage up very slowly



Step 2 As the voltage approaches the charging termination setting, the controller should start pulse charging. The controller is trying to pulse voltage into the batteries.

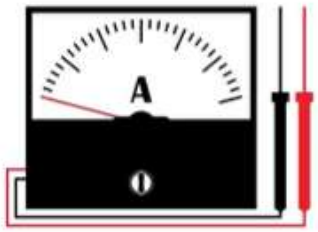


Step 3 The charging LED should be flashing on and off. If the controller has a fully charged LED, it should be off.





Testing Procedure for Multistage Charge Controller

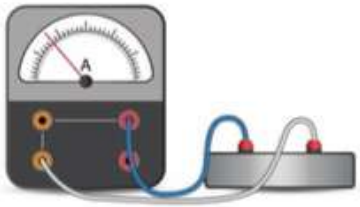






Voltmeter

Testing Procedure for Multistage Charge Controller



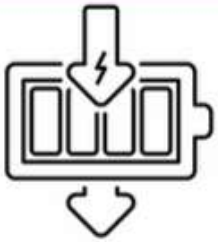
Step 1: Set the ammeter for the highest setting first, then change the settings downward. During the test, the current flow will range from amps to milliamps.



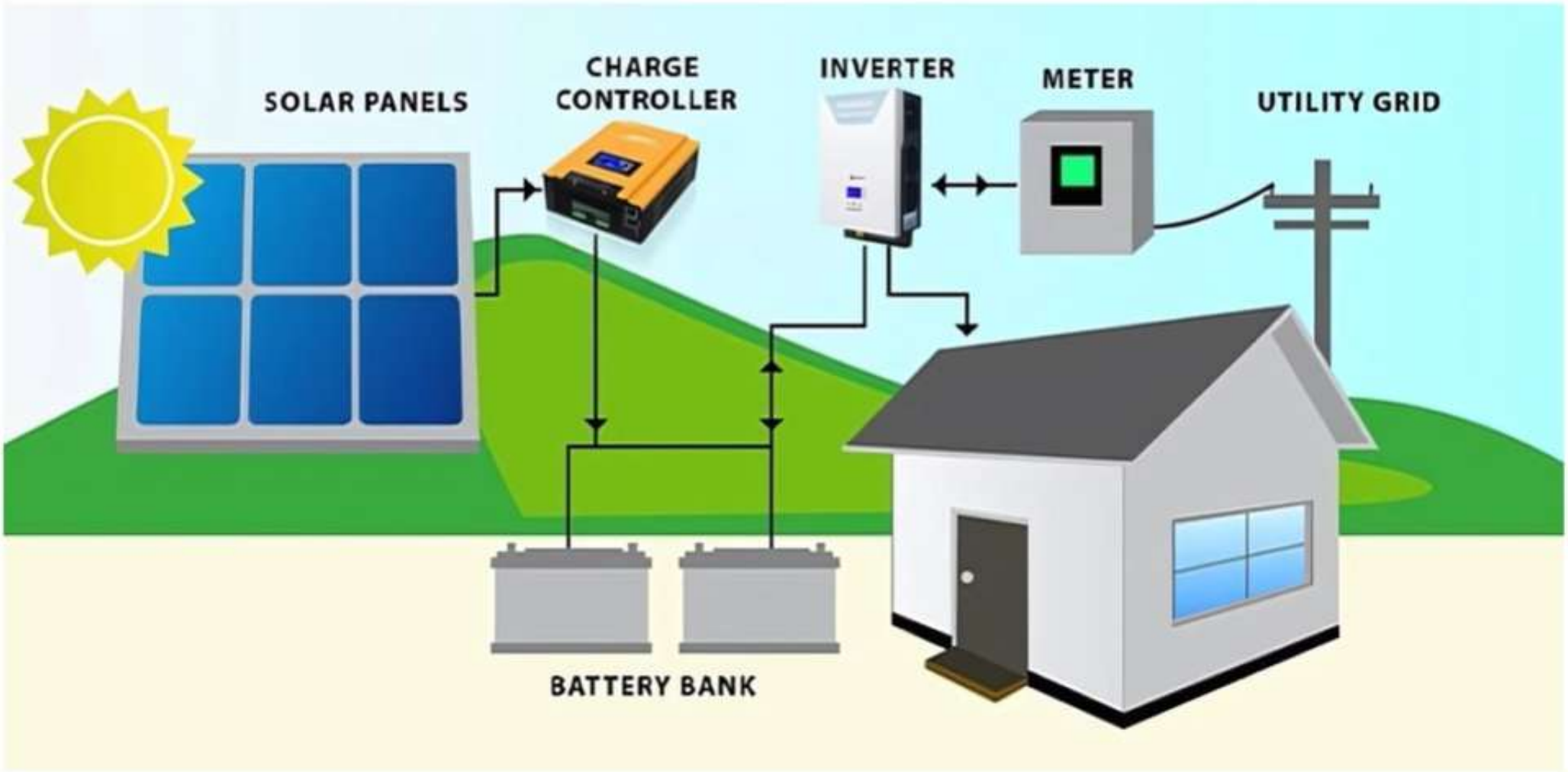
Step 2: Connect the ammeter to the controller's "battery" terminals



Step 3: Follow the same steps as described in testing procedure for series charge controller.



Step 4: The only modification will be to turn the power supply voltage up very slowly. At first, the current flow should be a few amps. As the voltage approaches the charging termination setting, the controller should start "trickle" charging at a few hundred milliamps.


SOLAR PANELS
CHARGE CONTROLLER
INVERTER
METER
UTILITY GRID
BATTERY BANK

Unit 6.6: Testing of batteries

Determining Conditions and Charge of Battery



General Conditions of Battery

1. The tops of the batteries should be clean and dry. Caps should all be in place and secure.

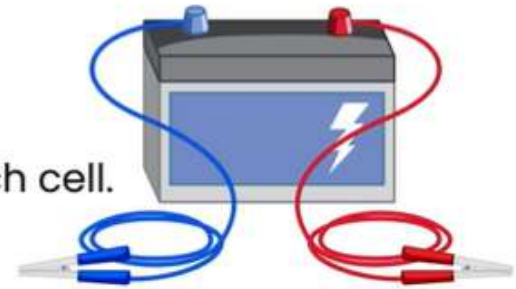
2. All wiring connections should be secure.

3. Confirm that there are no shelves, hooks or hangers above the batteries.

4. Check the electrolyte level of every cell in every non-sealed battery. It should always be above the top of the plates, but below the top of the battery cases.

5. Ventilation systems must be functional.

6. Label each battery with a number for the battery and numbers for each cell.



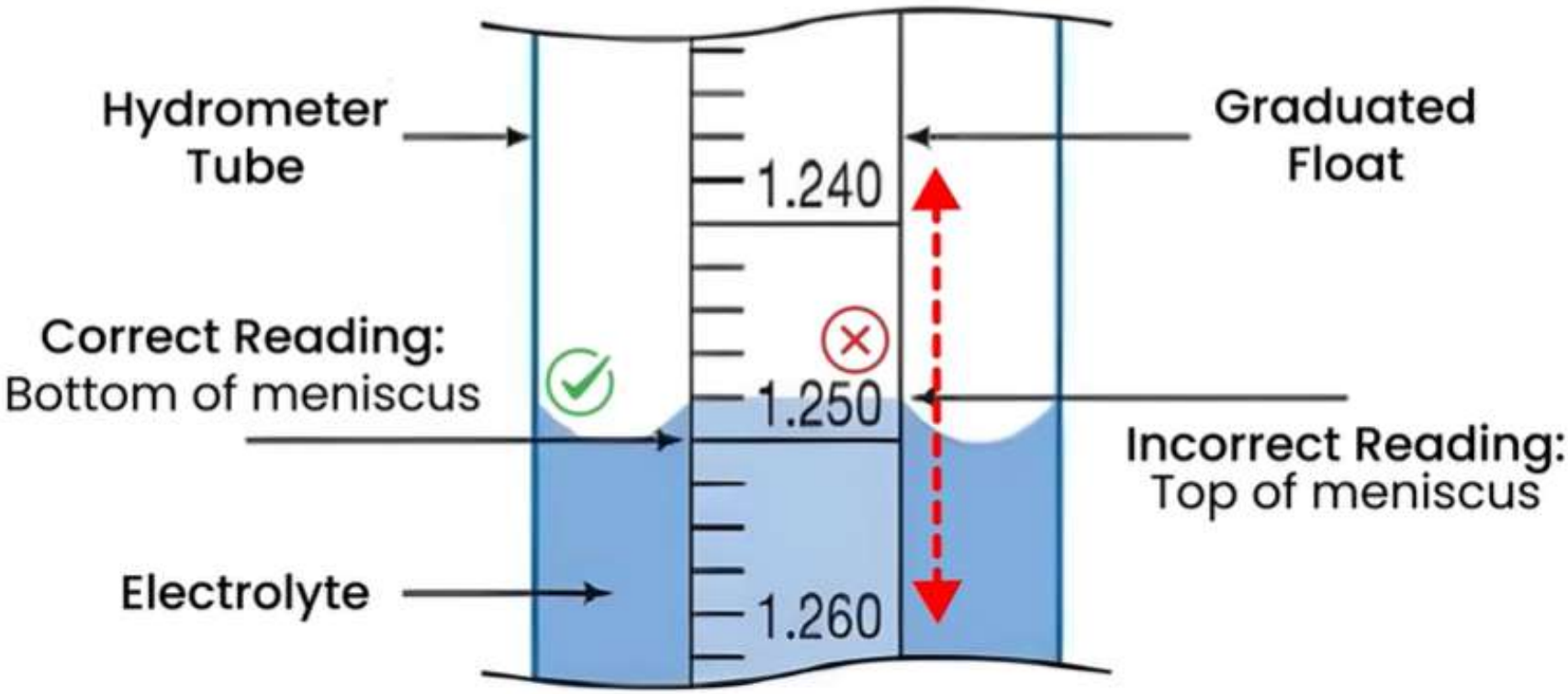


A **hydrometer** describes the state of charge by determining the specific gravity of the electrolyte.

Specific gravity is a measurement of the density of the electrolyte compared to the density of water.

Usually, the specific gravity of electrolyte is between 1.120 and 1.265. At 1.120, the battery is fully discharged. At 1.265, it is fully charged.

Reading a Hydrometer



Measurement of State of Charge Using a Hydrometer



Measurement of State of Charge Using a Hydrometer




0:06 / 0:08

Points to Remember When Using a Hydrometer

1. To use a hydrometer, squeeze the bulb while the inlet tube is still above the electrolyte level. Lower the hydrometer into the electrolyte and gradually release the bulb to suck in the electrolyte.

2. When emptying the hydrometer, slowly squeeze the bulb with the inlet just above the electrolyte level, but still inside the battery cell. These methods reduce the chances that electrolytes will spurt out of the battery or the hydrometer.

3. At the first cell being checked, fill and drain the hydrometer with electrolyte three times before pulling out a sample. This brings the hydrometer to the same temperature as the electrolyte.

4. Take a sample of electrolyte and allow the bulb to completely expand. Hold the hydrometer straight up and down, so the float is not touching the sides, top, or bottom of the tube.



5. Look straight across the electrolyte level to read the float. Ignore the curve of electrolyte up onto the sides of the hydrometer.



6. Record the specific gravity of each cell on the inspection sheet.

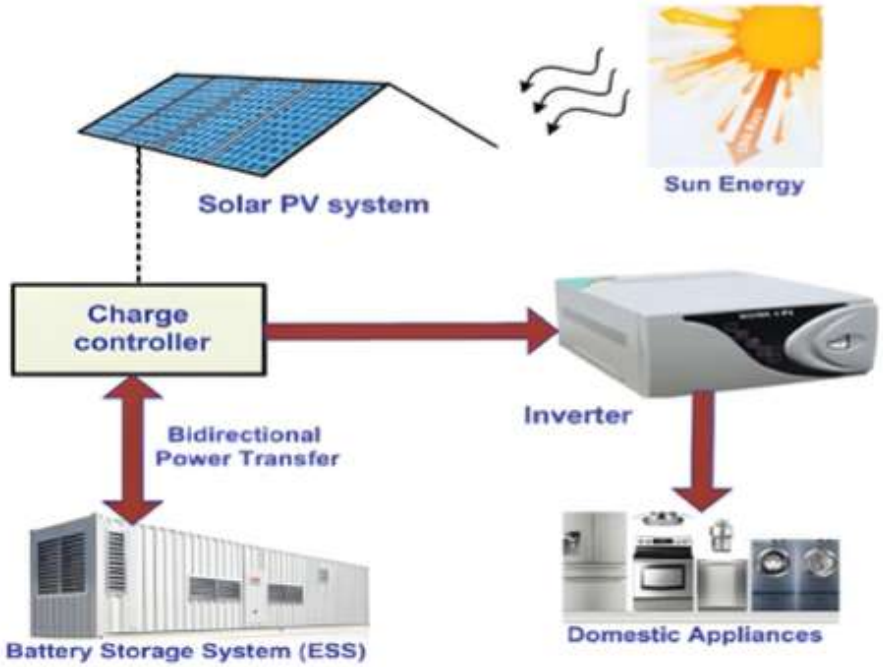
Specific Gravity Reading and State of Charge

Electrolyte Temperature (°C)	SG Reading at 100% SOC	SG Reading at 75% SOC	SG Reading at 50% SOC	SG Reading at 25% SOC	SG Reading at 0% SOC
48.9	1.249	1.209	1.174	1.139	1.104
43.3	1.253	1.213	1.178	1.143	1.106
37.8	1.257	1.217	1.182	1.147	1.112
32.2	1.261	1.221	1.186	1.151	1.116
26.7	1.265	1.225	1.19	1.155	1.12
21.1	1.269	1.229	1.194	1.159	1.124
15.6	1.273	1.233	1.198	1.163	1.128
10	1.277	1.237	1.202	1.167	1.132
4.4	1.281	1.241	1.206	1.171	1.136
-1.1	1.285	1.245	1.21	1.175	1.14
-6.7	1.289	1.249	1.214	1.179	1.144
-12.2	1.293	1.253	1.218	1.183	1.148
-17.8	1.297	1.257	1.222	1.187	1.152

Unit 6.7: Start the System


Steps to Start-up
Procedure of Solar
PV system

Let's Discuss the Start-up procedures of:



Standalone System



Grid Tied System

Start-up of Standalone System

Step 1



Start the inverter and make sure the inverter is coming on by turning on an AC load.

Step 2



Measure and record the current draw of the inverter in both idling and operating states.

Step 3

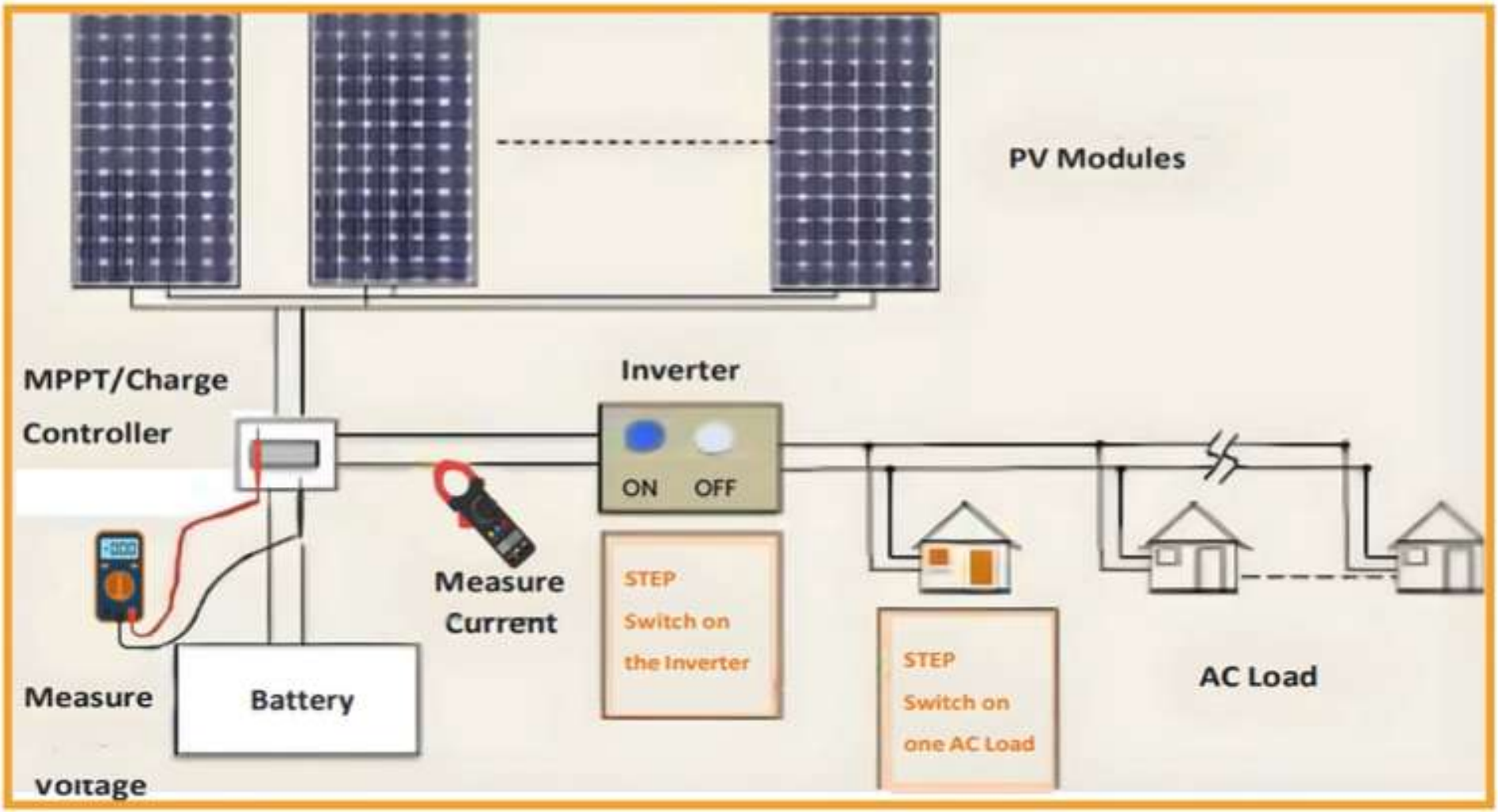


Measure and record the voltage drop between the inverter and battery on the positive and negative leg while under load.

Step 4



Measure the current drawn simultaneously and calculate the resistance to arrive at the loss between the battery and the inverter.


PV Modules
MPPT/Charge
Controller
Inverter
ON
OFF
Measure
Current
STEP
Switch on
the Inverter
STEP
Switch on
one AC Load
AC Load
Measure
Battery
voltage

Startup of Grid Connected System



Step 1: If inverter has an on/off switch ensure that it is in the off position .



Step 2: After you have completed all other tests, refer to the inverter's system manual and follow the start-up procedure.



Step 3: Typically, this will involve turning on the PV array DC isolator followed by the inverter AC isolator

If you do not have a display meter, use a clamp-on ammeter to measure either the AC or DC current.





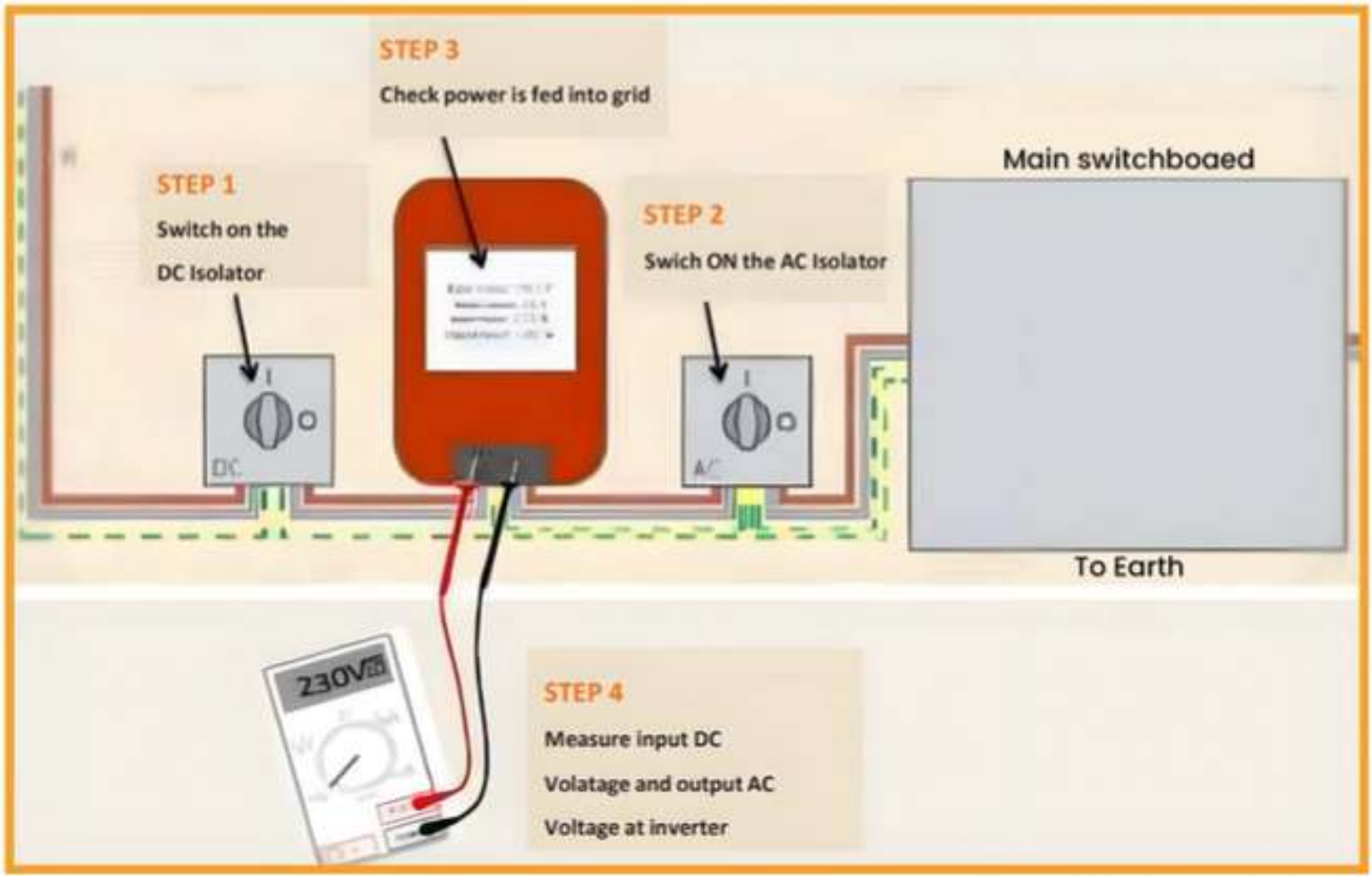
Step 4: Check and confirm that the solar array is feeding power onto the grid.



Step 5: Measure the AC output voltage.



Step 6: : If a kWh meter exists in the system, confirm that the inverter is producing the expected power output with respect to available DC power.


STEP 3
Check power is fed into grid
Main switchboaed
STEP 1
Switch on the
DC Isolator
STEP 2
Swich ON the AC Isolator
To Earth
230V
STEP 4
Measure input DC
Volatage and output AC
Voltage at inverter

Unit 6.8: Unintentional islanding Functionality Test

9.8.1 Unintentional Islanding Functionality Test

Unintentional islanding in a solar PV system occurs when a portion of the electrical grid, powered by solar panels, continues to operate in isolation from the rest of the grid during a power outage.



Unintentional Islanding situation can be dangerous for several reasons:

Safety Risks to Utility Workers

Workers may assume that the lines are de-energized and safe to work on when they are still being powered by the solar PV system.

Equipment Damage

The electrical equipment may be damaged due to mismatched load and generation.

Power Quality Issues

The isolated grid may experience voltage and frequency fluctuations, leading to power quality problems.

Please remember!

Conduct this test after the system has been operating correctly for a few minutes.

If there is more than one inverter, tests should be carried out for each inverter.

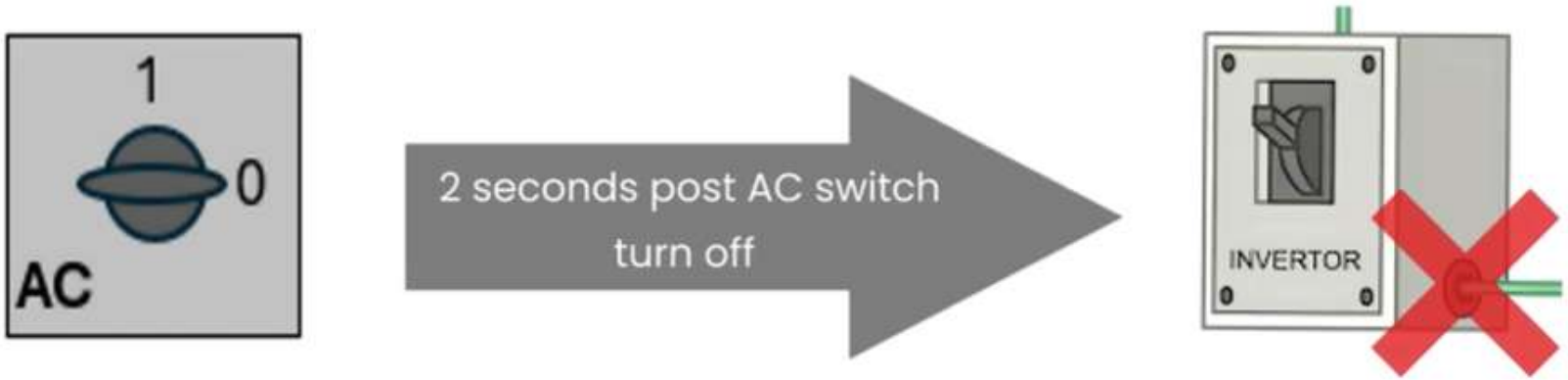
This test must be conducted during noon time in a sunny day.

PV system shall produce more than 20% of the rated output of the PV array or the inverter –whichever is less.

Performing Unintentional Islanding Functionality Tests



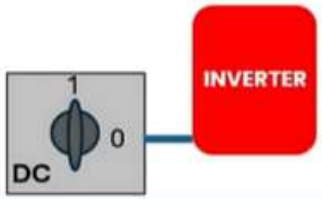
Test 1



Inverter Must Cease Supplying Power Within Two Seconds of a Loss of Mains

Inverter Must Cease Supplying Power Within Two Seconds of a Loss of Mains

Test 1



STEP 1 : Keep DC supply from the solar array connected to the inverter.



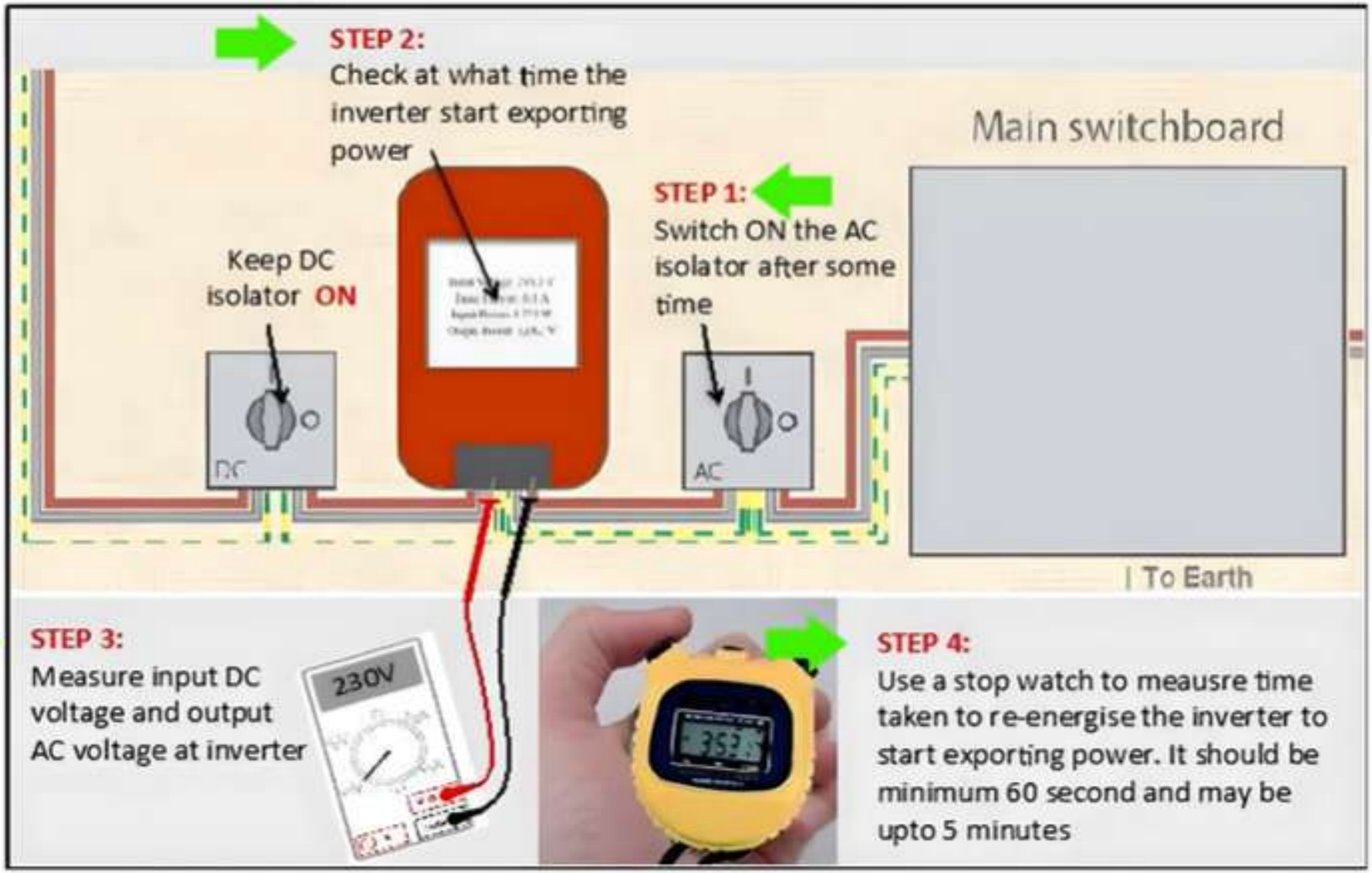
STEP 2 : Place the voltage probe in the inverter side of the AC main switch.



STEP 3 : Turn off the AC main switch through which inverter is connected to grid.



STEP 4 : Measure the time taken for the inverter to cease attempting to export power with a timing device and record.



Unintentional Islanding Functionality Test 1

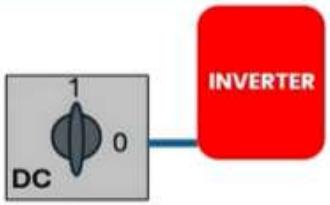
Test 2



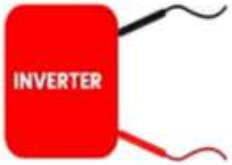
Inverter Must not Resume Supplying Power Until Main Switch is ON for More than 60 Seconds

Inverter Must Cease Supplying Power Until Main Switch is ON for More than 60 seconds

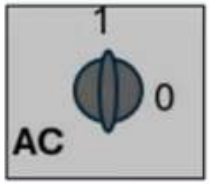
Test 2



STEP 1 : Keep DC supply from the solar array connected to the inverter.



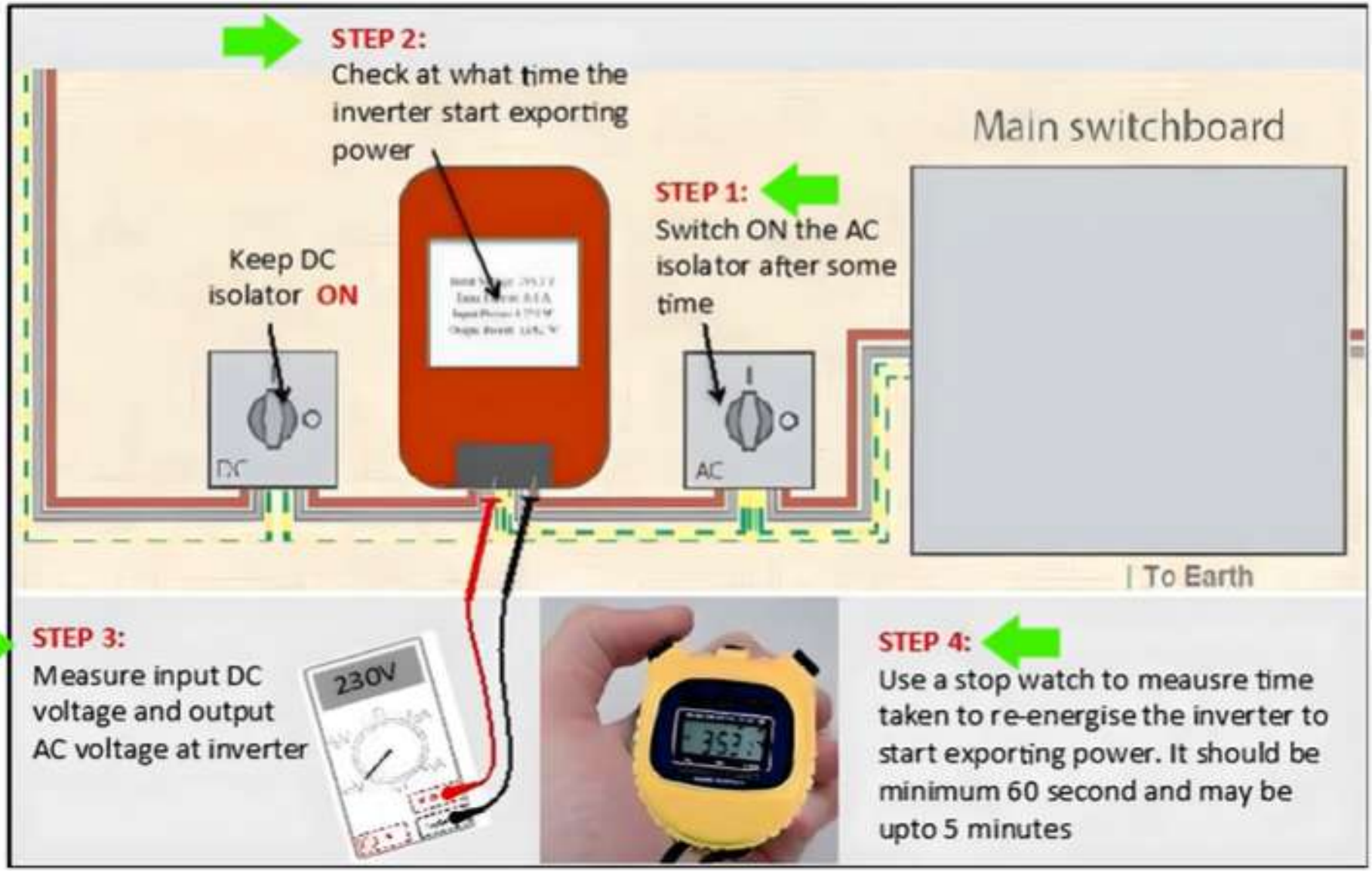
STEP 2 : Place the current probe in the inverter side of the AC main switch.



STEP 3 : Turn on the AC main switch through which inverter is connected to grid.



STEP 4 : Measure the time taken for the inverter to re-energise and start export power with a timing device and record.


STEP 2:
Check at what time the inverter start exporting power
Main switchboard
STEP 1:
Switch ON the AC isolator after some time
Keep DC isolator ON
DC
AC
To Earth
STEP 3:
Measure input DC voltage and output AC voltage at inverter
230V
STEP 4:
Use a stop watch to meausre time taken to re-energise the inverter to start exporting power. It should be minimum 60 second and may be upto 5 minutes

Unit 6.8: Inspection record sheet

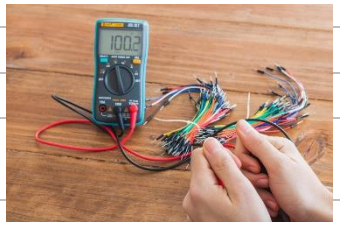
9.9.1 Sample Inspection Record Sheets







Overall System Inspection	
Array Location:	
1. Is the PV array free from shadow in all days of the year?	
3. Is the PV array protected from animals and vandalism? 4. Is the PV array having ample space for air-cooling?	
Equipment Location:	
	
5. Is the access to inverter battery and other equipment controlled? 6. Is the access to disconnect switch or circuit breaker are controlled?	
Battery Location:	
6. Is the access to disconnect switch or circuit breaker are controlled? 7. Is battery bank located in a dry location?	
8. Are batteries being in contact with cold surfaces such as concrete?	
9. Are batteries are placed in a box/ rack or inside a room?	
Testing of Solar Arrays:	
10. Is there any physical damages to any PV module?	
11. Is there any corrosion in the structure?	
12. Is there any spot shading on any module during noon time?	
13. Is there any loose fasteners or connections to the mounting surface?	
14. Are all conduit and connections tight and undamaged?	
15. Are all conduits and wire insulations undamaged?	
16. Are all junction boxes physically intact and tight?	
17. Did all wiring pass the pulling tests?	
18. Is there any wire that is readily accessible to any person?	
19. Did all individual string pass the continuity test?	
20. Did all individual string pass the open circuit voltage test?	
21. Did all individual string pass the polarity test?	

23. Is polarity in the junction box correct?	
24. Did the cable from PV array to the DC isolator pass continuity test?	
Wiring and earthing continuity tests:	
25. Did all cables pass continuity tests?	
26. Did earthing cable pass continuity tests?	
27. Is all metallic part in the system bonded correctly?	
28. Did all system components and wiring pass polarity tests?	
29. Is the resistance test using megger successful?	
Testing of Charge Controllers:	
30. Did you check the charge controller as per manufacturer's procedure?	
31. Did you check all terminals and wires for loose connection in CR?	
32. Did you check functionality and display of and indicators of CR?	
Testing of Batteries:	
33. Are batteries kept dry and vents are clear and clean?	
34. Are all connections in battery bank secured?	
35. Is electrolyte level of the batteries appropriate?	
36. Is ventilation system in battery bank room working?	
37. Did you level all battery with specific number?	
38. Did you note state of charge of all the cells of the batteries?	
Unintentional islanding functionality test:	
39. Did the system pass the TEST 1?	
40. Did the system pass the TEST 2?	